Computer & Communication Technology

HISTORY OF COMPUTERS

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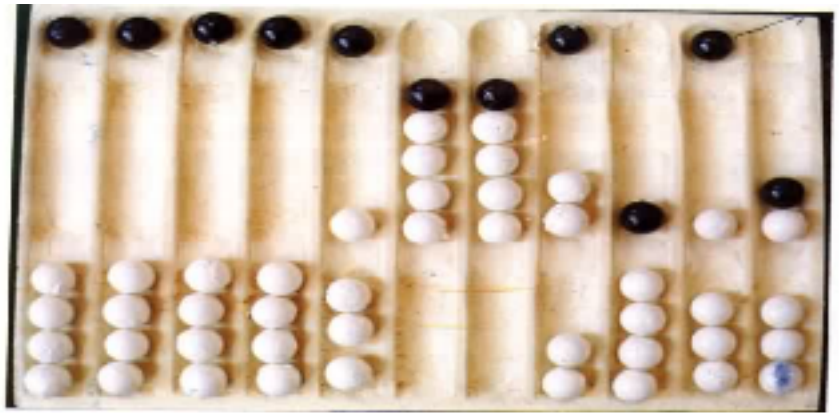
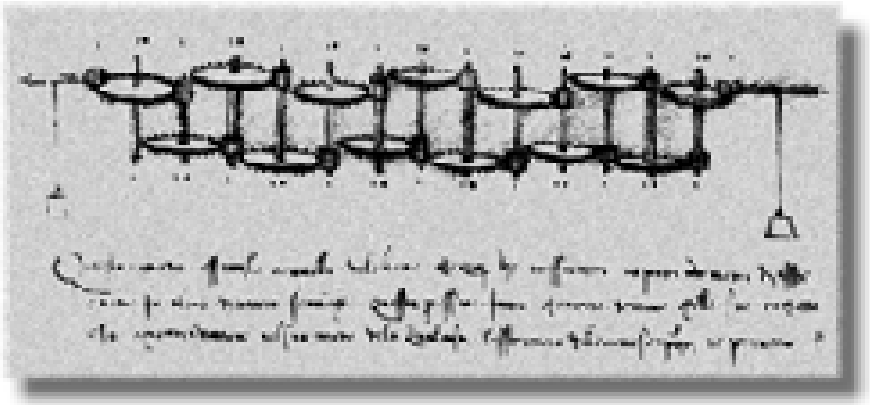
- Abacus (around 500 BC)
  - ***Abacus*** was an early aid for mathematical calculations.
  - A calculator used to add, subtract, multiply and divide by the Chinese
    - Oldest Abacus was used by the Babylonians in 300 BC

  - Still used in some parts of Asia

- Its only value is that it helps the memory of the human performing the calculation.
- A skilled abacus operator can work on addition and subtraction problems at the speed of a person equipped with a hand calculator (multiplication and division are slower).



Leonardo da Vinci (1452-1519)

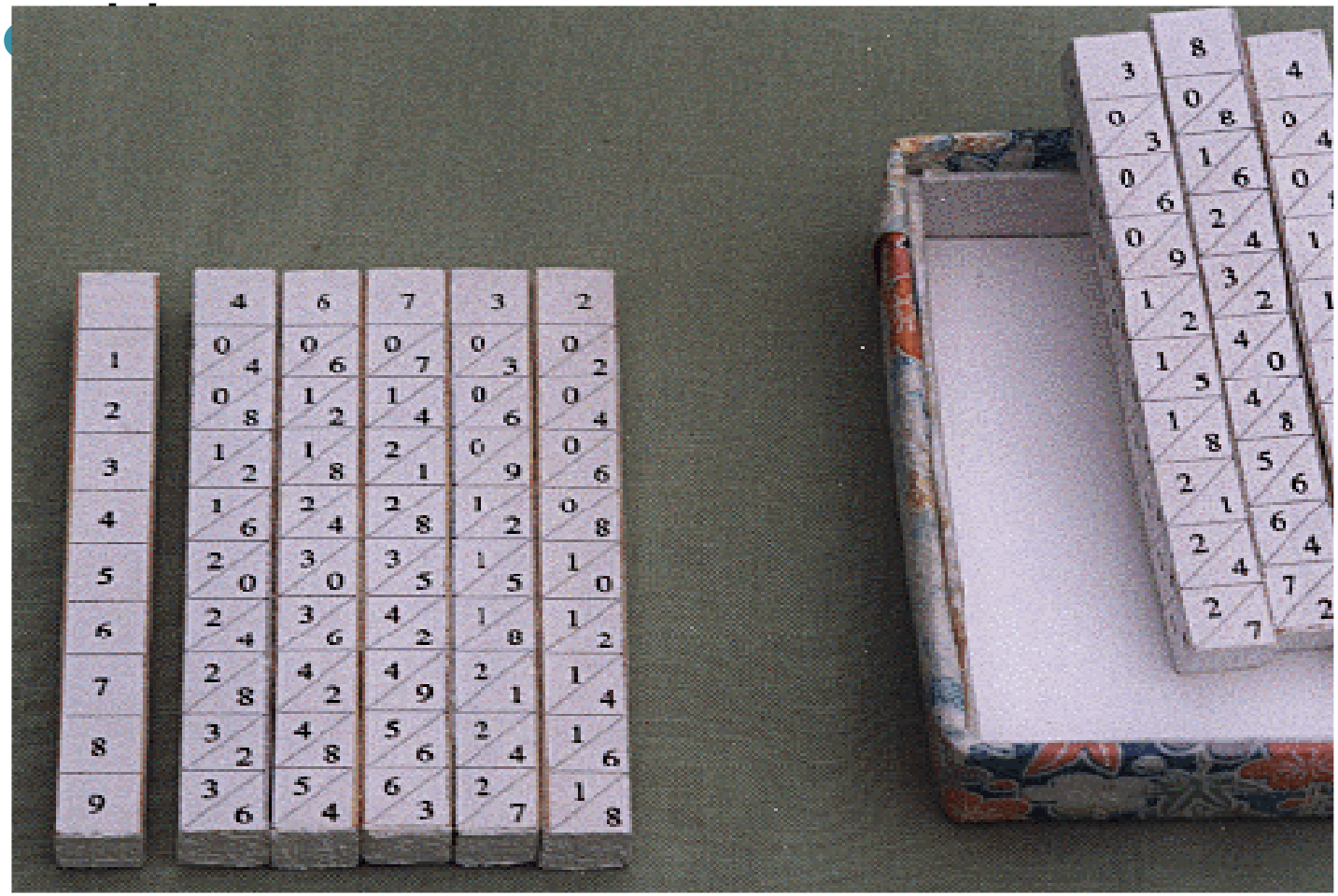


- Leonardo da Vinci (1452-1519) made drawings of gear-driven calculating machines but apparently never built any.

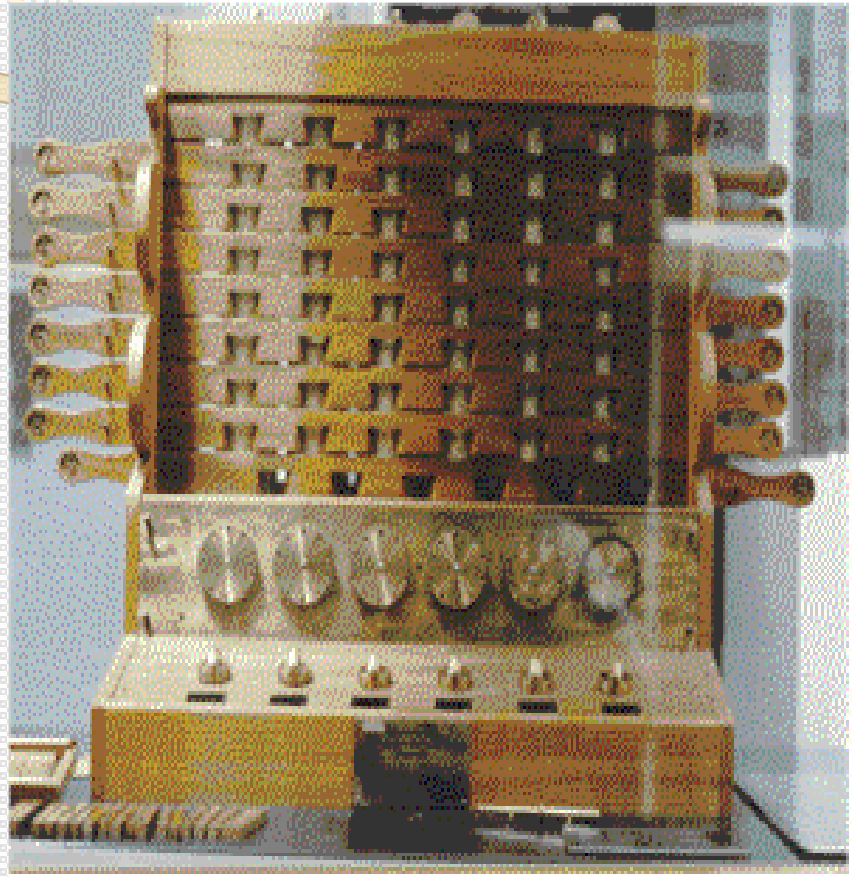
John Napier - 1617

- A Scotsman named John Napier invented ***logarithms***, which are a technology that allows multiplication to be performed via addition.
- Logarithm of each operand was originally obtained from a printed table.
  - Ex: $\log_2 x = 5$
- Logarithms are laid out in logarithmic tables, popularly known as 'log tables'.
- Used everywhere until1970s, when electronic calculators were invented.
- Napier also invented an alternative to tables, where logarithm values were carved on ivory sticks which are now called ***Napier's Bones***.

Napier’s Bones



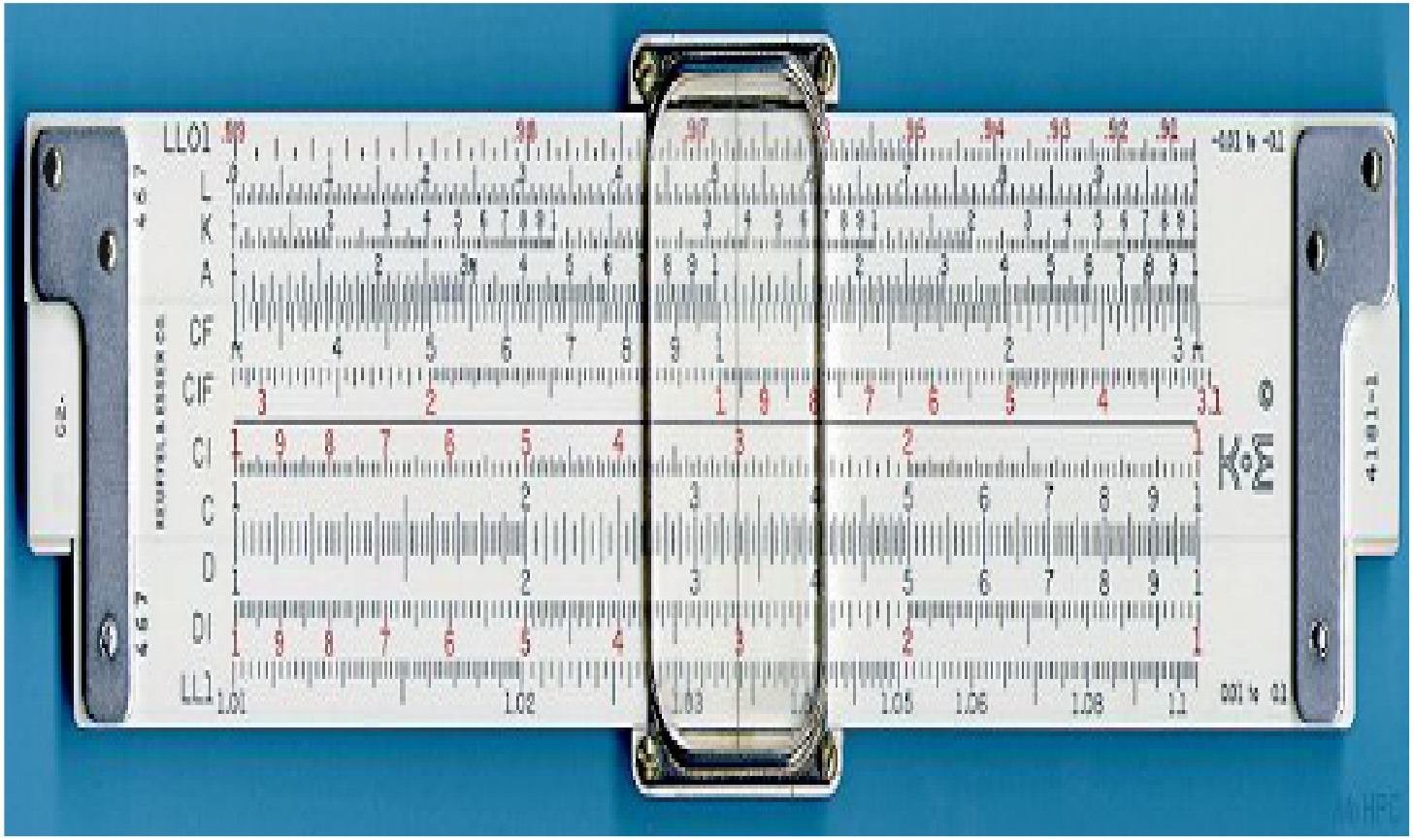
Calculating Clock



- The first gear-driven calculating machine to be built was probably the ***calculating clock***,
- Named by its inventor, the German professor Wilhelm Schickard in 1623.
- This device got little publicity because Schickard died soon afterward in the bubonic plague.

The slide rule 1630 - 1632

- Based on Napier’s rule of logarithms
- Napier's invention led directly to the ***Slide rule***,
- First built in England in 1632 and still in use in the 1960's by the NASA engineers of the Mercury, Gemini, and Apollo programs which landed men on the moon.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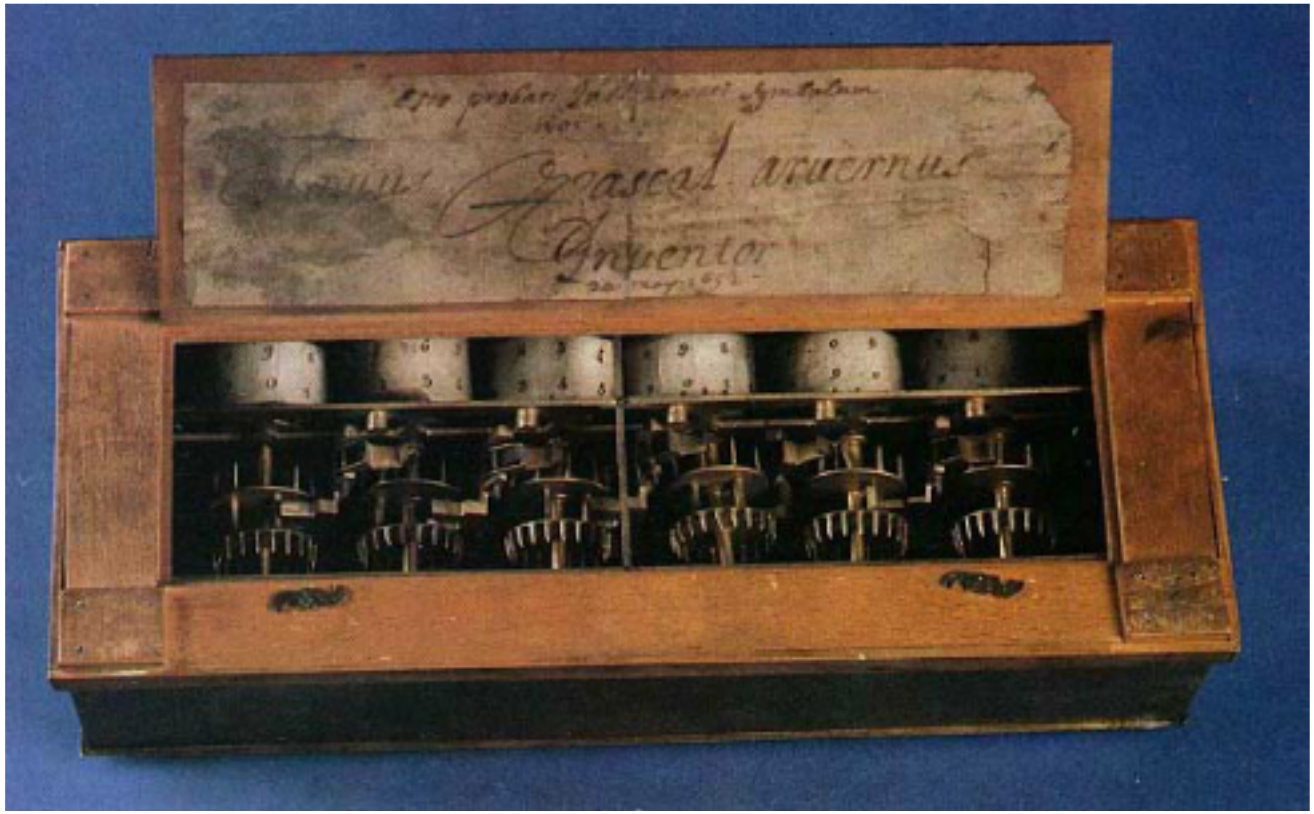
Pascaline (1642)

- Invented by Blaise Pascal, at age 19 as an aid for his father who was a tax collector
- A gear-driven one-function calculator (it could only add)
- Could not sell many because of their exorbitant cost and were not that accurate.

- Odometer portion of a car's speedometer used the very same mechanism as the Pascaline



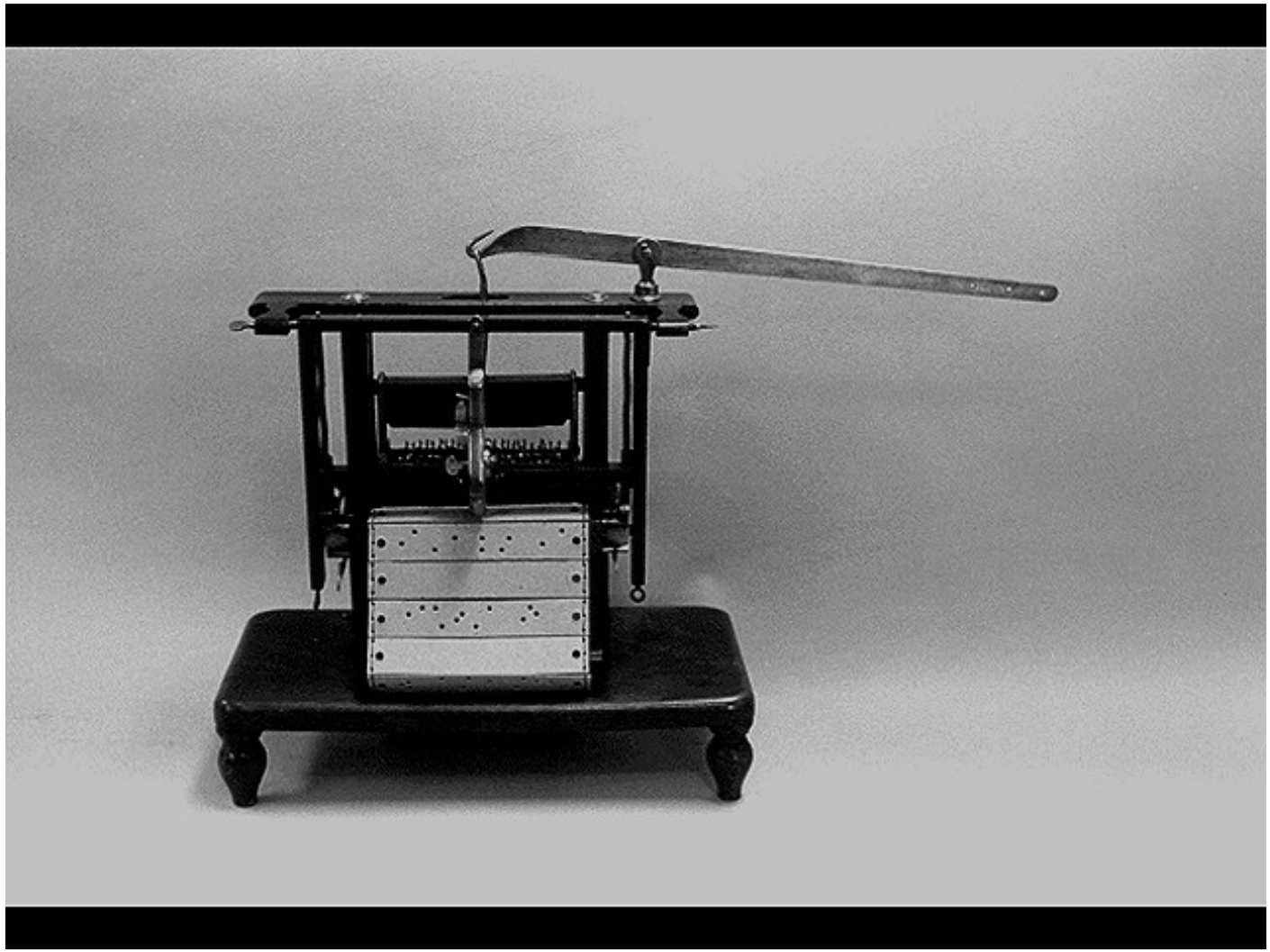
Stepped reckoner

- A four-function (addition, subtraction, multiplication, and division) calculator
- Built by the German, Gottfried Wilhelm Leibniz
  - (co-inventor with Newton of calculus)
- Leibniz was the first to advocate use of binary number system which is fundamental to the operation of modern computers.
- Leibniz is considered one of the greatest of the philosophers.

Joseph Marie Jacquard (1801)

- Invented a power loom that could base its weave upon a pattern read from punched wooden cards, held together in a long row by rope
- By selecting particular cards for Jacquard's loom you defined the woven pattern
- Used metal cards with punched holes to guide weaving process
- First stored program - metal cards
- First computer manufacturing
- Still in use today!

Jacquard Loom (19th Century



Technology -vs- Jobs

- Jacquard's technology was a real boon to mill owners, but put many loom operators out of work.
- Angry mobs smashed Jacquard looms and once attacked Jacquard himself.
- History is full of examples of labour unrest following technological innovation yet most studies show that technology has actually increased the number of jobs.

Difference Engine 1822

- In 1822 Charles Babbage proposed a steam driven calculating machine the size of a room which he called the ***Difference Engine***
- machine would be able to compute tables of numbers, such as logarithm tables
- He obtained government funding for this project due to the importance of numeric tables in ocean navigation powered by steam.
- Its construction proved exceedingly difficult and the project soon became the most expensive government funded project
- Ten years later the device was still nowhere near complete and funding dried up
- The device was never finished

Babbage - Analytic Engine

- Babbage was not discouraged, and by then was on to his next brainstorm, which he called the ***Analytic Engine***.
- This device, large as a and powered by 6 steam engines,
- It was programmable, thanks to the punched card technology of Jacquard.
- Analytic Engine had a key function that distinguishes computers from calculators: the conditional statement.
- A conditional statement allows a program to achieve different results each time it is run.

Ada Byron

- Charles Babbage befriended ***Ada Byron***, the daughter of the famous poet Lord Byron.
- She was only 19 and was fascinated by Babbage's ideas
- She began fashioning programs for the Analytic Engine, although still unbuilt.
- Analytic Engine remained unbuilt.
  - (British government refused to get involved with this one)
- Ada earned her spot in history as the first computer programmer.
- Ada invented the subroutine and was the first to recognize the importance of looping in computer programming.

US Census

- The next breakthrough occurred in America.
- The U.S. Constitution states that a census should be taken of all U.S. citizens every 10 years in order to determine the representation of the states in Congress.
- The first census of 1790 required only 9 months, by 1880 U.S. population had grown so much that the count for the 1880 census took 7.5 years.
- Automation was clearly needed for the next census.
- Census bureau offered a prize for an inventor to help with the 1890 census and this prize was won by Herman Hollerith who invented ***Hollerith desk***.

Herman Hollerith

- An American, developed (around 1890) the first electrically driven device called ***Hollerith desk***.
- It utilised punched cards and metal rods which passed through the holes to close an electrical circuit and thus cause a counter to advance.
- This machine was able to complete the calculation of the 1890 U.S. census in 3 years compared with 7 1/2 years for the 1880 census which was manually counted.

Hollerith Desk

- Hollerith's technique was successful and the 1890 census was completed in only 3 years at a savings of 5 million dollars.
- Hollerith built a company, the Tabulating Machine Company which eventually became International Business Machines, known today as *IBM*.

Mark I

- Also called Harvard ***Mark I*** computer which was built as a partnership between Harvard and IBM in 1944.
  - Howard Aiken of Harvard University and Thomas Watson of IBM.
- This was the first programmable digital computer made in the U.S.
- But it was not a purely electronic computer.
- Instead Mark I was constructed out of switches, relays, rotating shafts, and clutches.
- Mark I ran non-stop for 15 years, sounding like a room full of ladies knitting.

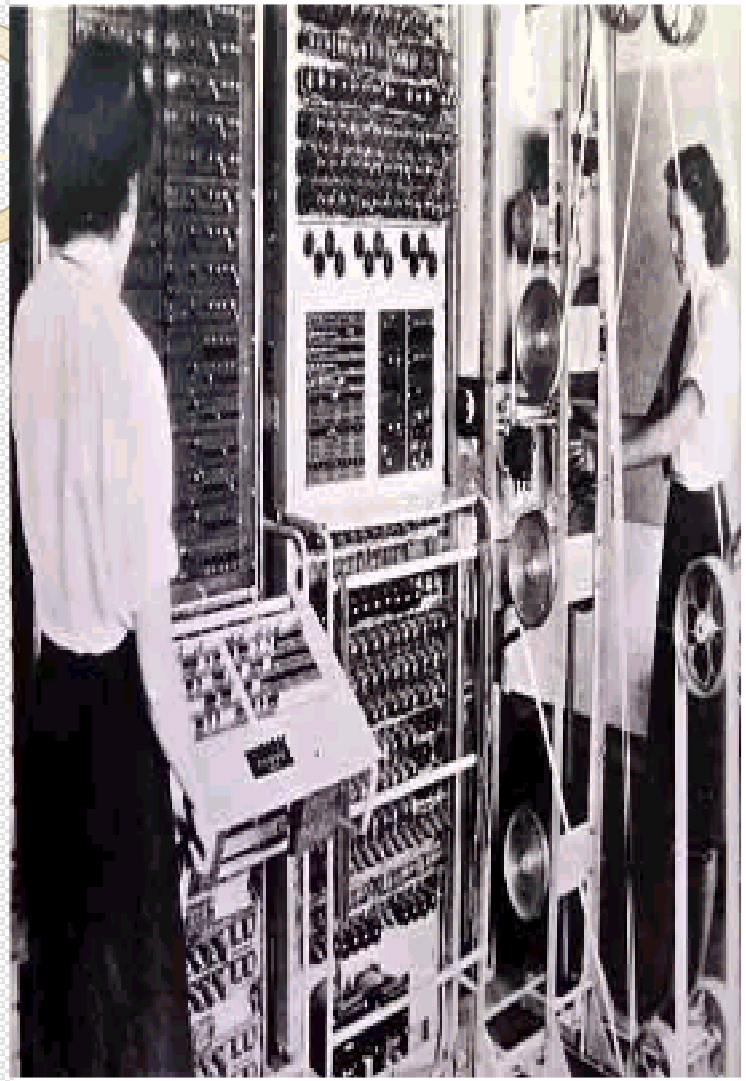
The First Bug

- One of the main programmers for Mark I was a woman, ***Grace Hopper***.
- Hopper found the first computer "bug": a dead moth that had gotten into the Mark I
- The word "bug" had been used to describe a defect.
- Hopper is credited with coining the word "debugging" to describe the work to eliminate program faults.

Atanasoff – Berry Computer

- Around 1937 John Atanasoff of Iowa State University and his assistant Clifford Berry built the first digital computer that worked electronically, called the ABC (Atanasoff-Berry Computer).
- This machine was basically a small calculator.
- This machine was the first to store data as a charge on a capacitor that is how today's computers store information in their main memory.
- It was also the first to employ binary arithmetic.

Colossus



- ***The Colossus***, was built during World War II by Britain for the purpose of breaking the secret codes used by Germany.
- Britain led the world in designing and building electronic machines dedicated to code breaking, and was able to read coded Germany radio transmissions.
- Not a general purpose, reprogrammable machine.

ENIAC (Electronic Numerical Integrator And Computer)

- John Mauchly and J. Presper Eckert of University of Pennsylvania developed a huge machine consisting of 18,000 vacuum tubes.
- ENIAC (Electronic Numerical Integrator And Computer) was born in 1946.
- It was a huge machine with a huge power requirement and two major disadvantages.
  - Maintenance was extremely difficult as the tubes broke down regularly and had to be replaced,
  - and also there was a big problem with overheating.

EDVAC (**E**lectronic **D**iscrete **V**ariable **A**utomatic **C**omputer)

- In the late 1940's John von Neumann (at the time a special consultant to the ENIAC team) developed the EDVAC (**E**lectronic **D**iscrete **V**ariable **A**utomatic **C**omputer) which pioneered the "stored program concept".
- This allowed programs to be read into the computer and so gave birth to the age of **general-purpose** computers.

UNIVAC - Universal Automatic Computer

- The first commercial computer produced in the United States.
- It was designed principally by J. Presper Eckert and John Mauchly, the inventors of the ENIAC.
- UNIVAC was also the first computer to employ magnetic tape.
- The first UNIVAC was accepted by the United States Census Bureau on March 31, 1951.

- The computer as we know it today had its beginning in the 19th century by a mathematician professor named Charles Babbage.
- He designed the analytical engine and its this design that the basic framework of computers of today are based on.
- Computers can be classified into generations.
- Each generation lasted for a certain period of time, and gave either a new or an improved computer or improvement to existing computer.